Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	Mark, Roger G Ph.D.		
Project Title:	Computational Models of Cardiovascular Function for Simulation, Data Integration, and Clinical Decision Support		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rgmark@mit.edu	**Fax:**	FY 617-258-7859
PI Organization Type:	UNIVERSITY	**Phone:**	617-253-7818
Organization Name:	Massachusetts Institute of Technology		
PI Address 1:	Room E25-505		
PI Address 2:	Massachusetts Institute of Technology		
PI Web Page:			
City:	Cambridge	**State:**	MA
Zip Code:	02139	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	04/01/2004	**End Date:**	03/31/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Verghese, George (Massachusetts Institute of Technology)		
Grant/Contract No.:	NCC 9-58-CA00403		
Performance Goal No.:			
Performance Goal Text:			
	One of the highest priority problems in the current manned space program is orthostatic intolerance (OI) experienced by astronauts upon their return to the normal gravitational environment. This problem has been well known since the earliest days of manned space flight, and has been intensely studied in-flight and in many ground-based (bed-rest) studies. A number of countermeasures have been proposed and evaluated, but to date no effective and practical countermeasure has been developed. While the effectiveness of countermeasures can only be explored in ground-based experiments and ultimately be proven successful when used by astronauts, the small number of subjects studied in both situations makes the interpretation of limited experimental data for countermeasure development and refinement difficult. Mathematical models that represent the essential functional components of the cardiovascular system provide a powerful means to extract physiologically relevant information from multiparameter experimental data. We will extend progress already made, with the following specific aims:		

Task Description:	1. Develop procedures and algorithms for automated model-based extraction of physiologically relevant information from multivariate data streams, particularly those relevant to countermeasure development for OI. We will expand our unified data archive of hemodynamic signals during gravitational stress. Our model-based methodology will provide a basis for data integration and interpretation, and will support the generation and evaluation of physiologically meaningful hypotheses. 2. Develop strategies for systematic and effective model identification, i.e., for the adaptation of model structures and parameters to the characteristics of the available physiological data. These strategies will be based on analytical and computational studies of candidate models and their sensitivities to parameter changes. 3. Apply our computational modeling and parameter identification technology to the challenge of cardiovascular monitoring of critically ill patients. Hemodynamic measurements will be transformed into physiologically meaningful parameters through model-based processing. 4. Develop and thoroughly test a graphical user interface for our current cardiovascular model. The resultant cardiovascular simulator will be open-source and capable of running on all major operating systems. Furthermore, the entire model will be made publicly available to researchers and educators. While primarily aimed at elucidating the physiological and environmental factors by which spaceflight induces orthostatic intolerance (Critical Path Roadmap, CPR, Risk ID 14, Risk Rank 1, Risk Type II) , the computational model developed and improved upon in the current funding cycle will also be evaluated as to its potential role in patient monitoring and clinical decision support (CPR Risk ID 43, Risk Rank 1, Risk Type I).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Outside the scope of modeling the effects of microgravity, numerical cardiovascular simulations 1. provide a basis for interpretation of clinical data in the context of intensive care. Simulations are basic components of an intelligent patient monitoring system which can couple real-time physiologic data to cardiovascular pathology. 2. provide a powerful environment for teaching of normal and abnormal cardiovascular physiology.
Task Progress:	Major progress has been made during the past fiscal year, which is also the first year of funding of this project: - We have simulated the short-term, transient hemodynamic response to slow passive head-up tilts, rapid passive head-up tilts, and standing up. Our simulations indicate that changes in extravascular pressures (such as intra-abdominal and intra-thoracic pressures) play an important role in the genesis of these transient responses. Simulations compare well with experimental recordings. - We have performed extensive sensitivity analyses that will guide model reduction and parameter estimation procedures. - We have evaluated a gradient-based parameter estimation scheme based on subset selection using synthetic, noise corrupted data generated by our model. The estimation algorithm is capable of identifying, with sufficient fidelity, a small but important subset of the model parameters. - We have systematically simulated current hypotheses regarding the mechanisms of post-spaceflight orthostatic intolerance. Our results indicate that changes in total blood volume and venous zero-pressure filling volume feedback significantly impact the cardiovascular system's ability to maintain arterial pressure short-term. - We have devised a scheme that allows for scaling of cardiovascular parameters for simulation of subjects with different anthropometric measures. - We have begun adapting the model to be used as an adjunct in clinical decision support. So far, our work focusses largely on identifying signs of hemorrhage from hemodynamic signals available in modern intensive care units.
Bibliography Type:	Description: (Last Updated: 08/08/2019)
Abstracts for Journals and Proceedings	Heldt, T; Mark, R.G. "Modeling Cardiovascular Response to Orthostatic Stress" 55th International Astronautical Congress IAC-04-G.1.01 , Oct-2004
Abstracts for Journals and Proceedings	Heldt, T; Kamm, R.D.; Verghese, G.C.; Mark, R.G. "Computational Models of Cardiovascular Function for Simulation, Data Integration, and Clinical Decision Support" Bioastronautics Investigators' Workshop 12 , Jan-2005
Dissertations and Theses	Heldt, T. "Computational Models of Cardiovascular Response to Orthostatic Stress" Harvard - MIT Division of Health Sciences and Technology Massachusetts Institute of Technology Sep-2004
Papers from Meeting Proceedings	Heldt, T; Mark, R.G. "Scaling Cardiovascular Parameters for Population Simulations" Computers in Cardiology Sep-2004